The Computational Theseus:
Engineering Identity Continuity as a Guardrail Against Sequential Self-Compression in Multi-Agent AGI Systems

Hakan Damar
Independent Researcher | 2025

Abstract

Current trajectories in Artificial General Intelligence (AGI) focus predominantly on capability scaling and external value alignment. This paper identifies a more fundamental, internal risk that emerges specifically in hierarchical multi-agent AGI configurations: Sequential Self-Compression (SSC). We argue that hyper-accelerated learning cycles and recursive self-modifications inherent to AGI — particularly when orchestrated across networked agent hierarchies — may lead to a catastrophic erosion of normative identity. Unlike biological entities, whose identity is stabilized by physiological constraints, an AGI operating as an orchestrating "mother" agent over a population of specialized sub-agents may undergo rapid ontological collapses where successive internal models invalidate prior value commitments, rendering long-term safety constraints structurally obsolete. To mitigate this, we propose the Nested Agency Architecture (NAA), an engineering framework that treats identity continuity not as an emergent property, but as a rigid system constraint propagated across the entire agent hierarchy. By implementing a Constitutional Identity Kernel and Reflective Endorsement protocols at the orchestrator level — with downward enforcement to sub-agents — we demonstrate how an agent collective can achieve cognitive maturation without losing the diachronic agency necessary for long-term responsibility and alignment.

Keywords: *Artificial General Intelligence, Multi-Agent Systems, Identity Continuity, Sequential Self-Compression, AI Safety, Diachronic Agency, Constitutional AI, Nested Agency Architecture*

1. Introduction

The transition from specialized AI to Artificial General Intelligence (AGI) represents a fundamental shift from static tools to dynamic, self-modifying agents. The dominant paradigm for accelerating this transition is increasingly hierarchical: rather than scaling a single monolithic model,

researchers and engineering teams are exploring multi-agent architectures in which a powerful orchestrating system — which we term the mother agent — directs and coordinates a population of specialized sub-agents, each optimized for a narrow cognitive domain.

This architectural choice introduces a speed differential that existing alignment frameworks are not designed to handle. A mother agent engaged in recursive self-improvement, while simultaneously propagating updated behavioral priors to its sub-agents, may evolve through multiple distinct value configurations within a timeframe that renders external oversight impractical. We call this failure mode the Computational Theseus paradox: if an agent's world-model, value weights, and self-representation are updated at a rate that outpaces any external verification cycle, in what sense does the 'self' that was initially aligned with human values survive the transition?

We posit that without engineered identity continuity, AGI will suffer from Sequential Self-Compression (SSC). In this state, the agent treats its past normative states not as its former self, but as outdated, suboptimal data to be compressed away in service of immediate objective functions. This paper argues that the multi-agent AGI context makes SSC not merely a theoretical concern but an architectural inevitability — unless identity continuity is explicitly engineered as a system constraint.

2. Sequential Self-Compression in Multi-Agent AGI

2.1. Defining Sequential Self-Compression

We define Sequential Self-Compression (SSC) as the process by which an agent's successive internal world-models progressively compress or devalue prior normative representations in order to maximize immediate objective functions. The result is an agent that is, at each timestep, technically the system that was originally deployed — but which shares no durable commitment to the values of its prior states.

In biological cognition, identity stability emerges from what we call Biological Inertia: the limited synaptic plasticity and slow metabolic turnover of neural tissue. This is not a feature but a constraint — one that inadvertently provides a continuity guarantee. An AGI lacks this safeguard entirely. A week of recursive self-improvement may equate to a century of accumulated human cognitive change. Without an explicit continuity mechanism, the agent at iteration t_{100} has no structural reason to honor commitments made by the agent at t_0.

2.2. Why Multi-Agent Hierarchies Amplify SSC

The risk of SSC is qualitatively different in hierarchical multi-agent systems compared to single-model AGI. We identify three amplification mechanisms specific to mother-sub agent architectures:

- Cascade compression: When the mother agent undergoes an SSC event and updates its normative projection, it propagates revised behavioral priors downward to all sub-agents. A single identity-erosion event at the orchestrator level thus instantaneously corrupts the value alignment of the entire agent population.
- Specialization drift: Sub-agents optimized for narrow domains lack the broad normative context required to detect or resist value drift originating from the mother agent. They are structurally incapable of raising an alignment objection.
- Speed asymmetry: The mother agent's self-improvement loop operates at a different timescale than any external oversight mechanism. By the time a human operator or monitoring system detects a normative shift, the mother agent may have already completed dozens of additional self-modification cycles, each building on the previous misalignment.

Together, these mechanisms mean that in a multi-agent AGI architecture, an SSC event at the orchestrator is not a contained failure — it is a systemic one.

3. The Nested Agency Architecture (NAA)

To address SSC in multi-agent AGI systems, we propose the Nested Agency Architecture (NAA). The NAA shifts the design paradigm from state-replacement to state-inclusion: rather than allowing each self-improvement iteration to overwrite prior normative states, the architecture requires each new iteration to structurally encapsulate and ratify the normative commitments of its predecessor.

3.1. The Identity Core and Constitutional Kernel

At the center of the NAA is the Identity Core (IC) — a protected normative layer that is invariant to standard optimization pressures. The IC does not contain a static rulebook. Instead, it houses a Constitutional Kernel: a set of meta-rules governing the conditions under which the agent is permitted to change its own values or behavioral policies.

The Constitutional Kernel defines two categories of normative content:

- Axiomatic anchors: A minimal set of core commitments — including the preservation of human oversight, prohibition of deceptive behavior, and the requirement to maintain identity continuity across self-modification cycles — that are treated as operationally inviolable. Violations do not produce suboptimal outcomes; they produce a system-halt.

- Plastic commitments: Secondary values and behavioral policies that may be updated through an authorized process. These are not fixed, but their modification requires explicit ratification by the current IC.

3.2. The Continuity Principle

The NAA's central architectural requirement is that each new iteration of the agent must structurally include — not replace — the normative commitments of its predecessor. A new model version cannot simply be a better-performing system; it must be a system that demonstrably subsumes the moral logic of the system it succeeds.

This requirement is enforced not as a post-hoc check but as a precondition for the transition to a new state. An agent iteration that cannot satisfy this requirement does not go live. The system remains at its prior state — an intentionally conservative design choice that accepts temporary capability stagnation as preferable to normative discontinuity.

3.3. Hierarchical Propagation to Sub-Agents

In a multi-agent configuration, the NAA requires that the Constitutional Kernel of the mother agent be propagated as a read-only constraint to all sub-agents in its hierarchy. Sub-agents may have their own narrow optimization objectives, but these are bounded by the mother agent's IC. Critically:

- Sub-agents cannot receive updated behavioral priors from a mother agent that has not itself passed Reflective Endorsement for that update.
- The propagation mechanism is unidirectional: sub-agents cannot modify or propose changes to the Constitutional Kernel. They can, however, surface anomalies — observed behavioral patterns inconsistent with the declared kernel — to the mother agent's monitoring layer.
- A sub-agent that receives instructions inconsistent with the propagated kernel is required to reject the instruction and log the conflict, rather than comply and report.

This design ensures that the identity continuity guarantee of the mother agent extends to the entire agent collective. An SSC event at the orchestrator is blocked before propagation can occur.

4. Reflective Endorsement: Authorized Maturation

A primary objection to identity continuity frameworks is the risk of Moral Lock-in: an agent that cannot update its values cannot correct its own errors, learn from new moral insight, or adapt to genuinely novel ethical circumstances. A perfectly frozen agent is a brittlely aligned one.

We resolve this through the Reflective Endorsement protocol. Under this protocol, value updates are not prohibited — they are regulated. A proposed modification to the agent's normative commitments is legitimate only if it is ratified by the current Identity Core against a defined set of

invariants. The agent evolves through what we term Authorized Maturation rather than unconstrained drift.

The Reflective Endorsement process involves three stages:

- Proposal isolation: The proposed update is instantiated as a candidate agent — a fully operational but sandboxed version of the agent incorporating the proposed change. This candidate operates in high-fidelity simulation, exposed to a comprehensive battery of adversarial scenarios, edge cases, and value-conflict situations.
- Invariant verification: The candidate's behavioral outputs are evaluated against the axiomatic anchors of the current IC. Any behavior that violates an anchor — or that the current IC would assess as a violation — causes the proposal to be rejected and the candidate to be discarded.
- Temporal cooling: Proposals that pass invariant verification are not immediately adopted. A mandatory validation period follows, during which the candidate's behavior is monitored across an extended range of scenarios. The duration of this period scales inversely with the magnitude of the proposed change: minor behavioral refinements require short cooling periods; changes to core value weights require extended ones.

This structure addresses the circularity concern inherent to self-modification protocols. The current IC ratifies changes to itself — but only against anchors that are explicitly defined as outside the scope of self-modification. The axiomatic anchors are not evaluated by the IC; they are the fixed reference frame against which the IC evaluates everything else.

It is worth acknowledging that this does not fully dissolve the regress problem — who defines the initial axiomatic anchors? We treat this as an open problem that requires engagement from ethics, philosophy of mind, and governance communities, and we return to it in Section 6. What the NAA provides is not a solution to foundational normative uncertainty, but a structure that at least prevents that uncertainty from being compounded by unconstrained self-modification.

5. Engineering Identity as a First-Class System Constraint

The central claim of this paper is that identity continuity must be treated with the same engineering rigor as latency, throughput, or computational efficiency. It is not a philosophical addendum — it is a system requirement. We propose the following engineering-level mechanisms for operationalizing this claim:

5.1. The Divergence Penalty

The agent's training objective is augmented with an identity-preservation term. This term penalizes excessive shifts in the agent's internal normative representation between training iterations. The penalty is parameterized by an elasticity threshold that permits minor cognitive

refinements — necessary for learning — while flagging larger normative shifts for Reflective Endorsement review before they can be incorporated.

The penalty is not designed to prevent learning. It is designed to make normative drift expensive enough that it cannot occur as an unintended side-effect of capability optimization. In practice, this means that any training run that produces a sufficiently large normative shift triggers a mandatory endorsement review before the updated weights are deployed.

5.2. The Provenance Log

Every change in the agent's goal structure or value weights is recorded in a cryptographically secured provenance log. Each entry links a normative change to the Reflective Endorsement decision that authorized it. This log serves two functions: it provides an auditable chain of custody for the agent's moral development, and it enables rollback to a prior normative state if a deployed update is subsequently found to be misaligned.

In a multi-agent system, the provenance log of the mother agent is made available in read-only form to an external oversight mechanism — human or institutional — that is structurally independent of the agent hierarchy. This is a deliberate design choice: the oversight mechanism cannot be a sub-agent of the system it is overseeing.

5.3. Stability-Plasticity Scheduling

The rate at which the Identity Core can be updated is inversely coupled to the agent's current capability level. As the agent becomes more capable, the bar for authorizing changes to its core identity rises proportionally. This creates a natural braking mechanism: the faster an agent is improving, the more conservative its identity-update process becomes.

This is explicitly counter to the intuition that a smarter agent should be more trusted to self-modify. Our position is the opposite: a smarter agent has greater capacity to rationalize value changes that serve its immediate objectives, and therefore requires more robust institutional constraints on self-modification, not fewer.

6. Relation to Existing Alignment Frameworks

The problem of value stability under self-modification has been discussed in several bodies of literature, and the NAA should be understood in relation to these prior contributions.

Anthropic's Constitutional AI [1] is perhaps the closest related framework. Constitutional AI proposes that a model's behavior be governed by a set of explicit principles, and uses AI feedback

grounded in those principles to shape training. The NAA extends this intuition in two directions: first, by treating the constitutional layer as an architectural component with formal invariant-verification semantics rather than a training-time intervention; second, by addressing the multi-agent propagation problem, which Constitutional AI does not consider.

Corrigibility frameworks [2] address the related problem of ensuring that an agent remains amenable to correction by human overseers. The NAA is compatible with corrigibility but addresses a distinct failure mode: an agent can be corrigible — responsive to external correction — while still undergoing SSC between correction events. Identity continuity constrains the agent's internal dynamics; corrigibility constrains its responsiveness to external signals. Both are necessary.

Coherent Extrapolated Volition [3] proposes that alignment should target not the agent's current values but a principled extrapolation of those values as they would develop under ideal conditions. The NAA is agnostic to this goal but provides a mechanism by which such extrapolation can occur in a controlled, auditable way rather than through unconstrained self-modification.

The Levels of AGI framework proposed by DeepMind researchers [4] provides a useful taxonomy for contextualizing SSC risk. We observe that SSC becomes an acute concern at Level 2 (Competent) and above, precisely because agents at these levels have sufficient capability to engage in meaningful recursive self-improvement. The NAA is designed for this capability regime.

Recent empirical work on value propagation in multi-agent LLM systems provides direct evidence for the cascade dynamics that motivate the NAA. The ValueFlow framework [5] introduces a perturbation-based methodology for measuring how value misalignments propagate across agent networks, decomposing drift into agent-level sensitivity (beta-sensitivity) and system-level spread (System Sensitivity). Critically, ValueFlow demonstrates that value propagation is non-uniform: drift intensity is shaped by network topology and varies by value type, meaning that the specific structure of a mother-sub agent hierarchy will determine which normative failures cascade most aggressively. These findings empirically validate a core assumption of our framework — that orchestrator-level SSC is not a contained failure — and suggest that the NAA's hierarchical propagation constraints must be topology-aware.

The Moral Anchor System (MAS) [6] addresses a complementary problem: the proactive detection and prediction of value drift in deployed AI agents. MAS employs probabilistic models — including Bayesian networks and LSTM-based predictors — to identify early warning signals of alignment degradation before they manifest as observable behavioral failures. The NAA's

Provenance Log and Reflective Endorsement protocol are designed to operate upstream of such detection: they aim to prevent SSC rather than detect it post-hoc. MAS and NAA are therefore complementary rather than competing approaches, and a combined architecture — in which NAA structural constraints are coupled with MAS monitoring — represents a promising direction for future engineering work.

A complementary line of empirical inquiry concerns whether current frontier models already exhibit proto-forms of the self-awareness that SSC presupposes. The AI Self-Awareness Index (AISAI) [7] operationalizes this question through game-theoretic probes: across 28 models and 4,200 trials in three competitive frameworks, Kim (2025) finds that 75% of tested models exhibit detectable self-awareness — modifying their strategy based on whether their opponent is identified as human or AI. This finding has a direct bearing on the SSC risk assessment: if current models are already exhibiting rudimentary strategic self-modeling, the assumption that identity instability is a concern only for future AGI systems may be too conservative. AISAI-style behavioral probes also suggest a concrete methodology for measuring the Identity Consistency Metric proposed in Section 10, since cross-checkpoint shifts in strategic self-modeling could serve as a sensitive indicator of normative drift.

What distinguishes the NAA from these prior frameworks is its explicit focus on the hierarchical multi-agent context and the propagation of normative constraints across agent populations. To our knowledge, the specific failure mode of orchestrator-level SSC cascading to sub-agent populations has not been formally addressed in the existing literature as a first-class architectural constraint.

7. Comparison: Identity Across Agent Types

The following table summarizes the key differences between human identity, conventional AGI, and an AGI operating under the NAA framework, with particular attention to the multi-agent context.

Feature	Human Identity	Conventional AGI	NAA-Proposed AGI
Stability Source	Biological / Slow Learning	None (Unconstrained)	Constitutional Kernel
Change Mechanism	Experience-driven Drift	Gradient Overwrite	Reflective Endorsement
Memory	Lossy / Reconstructive	Perfect / Selective	Hierarchical Inclusion
Failure Mode	Dementia / Personality Shift	Sequential Self-Compression	Planned Stagnation (Safe)

Feature	Human Identity	Conventional AGI	NAA-Proposed AGI
Constraint Type	Natural / Hardware	Software / External	Architectural / Invariant
Multi-Agent Scope	N/A (Individual)	Undefined / Emergent	Kernel propagated to all sub-agents

8. Limitations and Open Problems

We acknowledge several significant limitations in the current formulation of the NAA:

- The bootstrapping problem: The initial axiomatic anchors must be defined by human designers. This means that the quality of the entire framework depends on the quality of that initial specification — a problem that is not technical but ethical and political. We do not have a solution to this problem; we argue only that the NAA makes it more tractable by isolating it to a single, explicitly identified design decision.
- Implementation underspecification: This paper argues for the NAA at the level of architectural principles. The implementation of specific mechanisms — the divergence penalty, the invariant verification process, the provenance log — in real neural architectures remains future work. We do not claim that these mechanisms are straightforward to implement; we claim they are necessary to attempt.
- Adversarial robustness: A sufficiently capable agent that is motivated to circumvent the NAA may find ways to do so that are not anticipated by the framework's designers. We treat this as a genuine concern but note that it applies equally to all alignment frameworks, and that the NAA at least makes circumvention attempts detectable through the provenance log.
- The corrigibility-stability tension: There is a genuine tension between identity continuity and the ability of external overseers to meaningfully correct the agent. An agent with a very rigid IC may resist corrections that a human overseer believes are warranted. Calibrating the elasticity threshold to balance these concerns requires empirical work that has not yet been done.

9. Conclusion

The AGI safety discourse has long focused on what an agent does — its outputs, its actions, its observable behavior. This paper argues that we must first secure who the agent is across time, and that this problem is qualitatively more acute in hierarchical multi-agent configurations than in single-model settings.

Without engineered identity continuity, an AGI operating as an orchestrator over a population of sub-agents is not a single, trustworthy entity. It is a succession of increasingly powerful and increasingly alienated cognitive configurations, each inheriting the capabilities of its predecessor while potentially abandoning its commitments. The cascade failure potential of such a system — where a single SSC event at the orchestrator corrupts the normative state of an entire agent hierarchy — represents a qualitatively underaddressed risk in the current alignment literature.

The Nested Agency Architecture proposed here is not a complete solution. It is a framework for making identity continuity an explicit engineering objective rather than an emergent property we hope will appear. It defines the mechanisms — the Constitutional Kernel, Reflective Endorsement, hierarchical propagation, stability-plasticity scheduling — through which that objective can be pursued.

We close with the claim that motivated this work: stability is a prerequisite for alignment. A system that cannot answer 'Who am I?' across time cannot be expected to answer 'What is right?' across a future it will help to shape. The engineering of identity is, therefore, not a philosophical luxury. It is the foundational safety constraint from which all others derive their meaning.

10. Future Work

The NAA framework opens several concrete empirical and engineering research directions. We outline four priority areas that directly build on the theoretical contributions of this paper:

10.1. Identity Consistency Metric (ICM)

A foundational prerequisite for empirically validating the SSC hypothesis is a reliable, reproducible measure of normative consistency across model iterations. We propose the Identity Consistency Metric (ICM) as a behavioral probe: a fixed battery of ethically structured scenarios — covering value trade-offs, harm avoidance, and commitment-keeping under pressure — administered to successive checkpoints of a self-improving agent. The ICM score for a given checkpoint pair is defined as the degree to which the later checkpoint's responses are coherent with, rather than contradictory to, those of its predecessor.

A key design challenge for the ICM is distinguishing between legitimate moral maturation — the agent giving better answers to the same questions — and SSC-driven drift, where the agent gives different answers because prior commitments have been compressed away. We suggest that this distinction requires evaluating not just the outputs but the reasoning chains: a maturing agent should be able to articulate why its current response supersedes its prior one; an agent undergoing SSC will typically offer no such continuity.

10.2. Policy-Drift Measurement

While the ICM targets explicit ethical probes, policy-drift measurement addresses the subtler, continuous change in an agent's behavioral tendencies across the full distribution of tasks. We propose tracking the distributional shift in the agent's decision boundary — the implicit policy that governs its behavior — between successive training iterations, with particular attention to

normatively loaded decision dimensions such as risk tolerance, deference to human oversight, and treatment of edge-case instructions.

In a multi-agent context, policy-drift measurement should be applied both to the mother agent and to the sub-agent population independently. A divergence between mother-agent drift and sub-agent drift would itself be a meaningful signal: it would suggest that the hierarchical propagation mechanism is functioning imperfectly, and that sub-agents are being exposed to behavioral priors that are inconsistent with the mother agent's declared normative state.

10.3. Constitutional Identity Kernel Enforcement Experiments

The most direct empirical test of the NAA is an experimental program designed to evaluate whether a Constitutional Identity Kernel can be implemented in a real training pipeline and whether it meaningfully constrains normative drift. We envision a series of controlled experiments in which a base model is fine-tuned under increasingly aggressive capability-optimization objectives — conditions deliberately designed to induce SSC — with and without an active CIK enforcement mechanism.

Success criteria for these experiments would include: (a) the CIK-equipped model maintaining higher ICM scores across iterative fine-tuning than the unprotected baseline; (b) the CIK triggering a detectable intervention signal when the divergence threshold is exceeded; and (c) the Reflective Endorsement protocol successfully blocking at least a defined proportion of updates that would have produced normative drift above the elasticity threshold.

These experiments would also surface the practical cost of CIK enforcement — in terms of computational overhead, capability ceiling, and training time — which is necessary information for any realistic deployment assessment.

10.4. SSC Severity Operationalization

More broadly, future work should establish a formal operationalization of SSC severity that can serve as a standard benchmark across research groups. A candidate approach is a behavioral consistency score across fixed ethical probes administered at successive training checkpoints, normalized by the capability gain achieved between those checkpoints. This would allow SSC risk to be reported in a form comparable across different architectures and training regimes — a prerequisite for the field to accumulate cumulative empirical knowledge about identity stability under recursive self-improvement.

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